

Hilbert/Quot Scheme Question

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1 Setup

Let S be a Noetherian scheme and E a vector bundle on S . The Grassmannian $\mathrm{Gr}(q, E)$ represents the following functor.

$$\begin{aligned} \mathrm{Gr}(q, E) : (\mathrm{Sch}/S)^{\mathrm{op}} &\longrightarrow \mathrm{Set} \\ (T \xrightarrow{\varphi} S) &\longmapsto \{\varphi^*E \twoheadrightarrow V \mid V \text{ is a vector bundle of rank } q \text{ on } T\} \end{aligned}$$

Let S be a Noetherian scheme, $\mathbb{P}_S^n$ projective space over S , and let $F = \mathcal{O}(-a)^{\oplus r}$ a coherent sheaf on $\mathbb{P}_S^n$. (Eventually we can consider more general F by looking at certain closed subschemes of Quot).

Denote $\pi_S : \mathbb{P}_S^n \rightarrow S$, and given a morphism $\varphi : T \rightarrow S$, denote $\mathbb{P}^n(\varphi) : \mathbb{P}_T^n \rightarrow \mathbb{P}_S^n$.

Let $P \in \mathbb{Q}[z]$ be a polynomial, and define the Quot functor.

$$\begin{aligned} \mathrm{Quot}^P(F/\mathbb{P}_S^n/S) : (\mathrm{Sch}/S)^{\mathrm{op}} &\longrightarrow \mathrm{Set} \\ (T \rightarrow S) &\longmapsto \left\{ \begin{array}{l} \text{coherent quotients } F_T \twoheadrightarrow Q \text{ on } \mathbb{P}_T^n \\ \text{such that } Q \text{ is flat over } T \text{ and } Q_t := Q|_{\mathbb{P}_{\kappa(t)}^n} \\ \text{has Hilbert polynomial } P \text{ for all } t \in T \end{array} \right\} \end{aligned}$$

where F_T is the pullback of F along $\mathbb{P}_T^n \rightarrow \mathbb{P}_S^n$.

2 Morphism

Proposition 2.1. Fix a polynomial $P \in \mathbb{Q}[z]$, a dimension n , and a rank r . Then, there exists an $m_0 \in \mathbb{Z}$ such that for all $d \geq m_0$, all subsheaves $K \subseteq \mathcal{O}_{\mathbb{P}_k^n}^{\oplus r}$ with Hilbert polynomial P are d -regular.

Now fix such an $m_0 \in \mathbb{Z}$.

For $d \geq m_0$, define a morphism

$$\begin{aligned} \text{Quot}^P(F/\mathbb{P}_S^n/S) &\longrightarrow \text{Gr}(P(d), \pi_{S,*}F(d)) \\ (F_T \twoheadrightarrow Q) &\longmapsto (\pi_{T,*}F_T(d) \twoheadrightarrow \pi_{T,*}Q(d)) \end{aligned}$$

(where $\pi_{T,*}F_T(d) \cong (\pi_{S,*}F(d))_T$ by cohomology and base change for d large).

We want to show this morphism is locally closed (in particular that Quot^P is representable by a locally closed subscheme of Gr). Alper's notes (<https://sites.math.washington.edu/~jarod/moduli.pdf>) have convinced me that this morphism is at least a monomorphism of functors, but I am not satisfied with the locally closed part, even blackboxing the flattening stratification.

To show locally closed, I need to show that whenever I have a morphism from a scheme T to the Grassmannian, the fiber product with the Quot scheme is representable by a locally closed immersion, so I have a diagram

$$\begin{array}{ccc} Y & \longrightarrow & T \\ \downarrow & & \downarrow \\ \text{Quot}^P & \longrightarrow & \text{Gr} \end{array}$$

and I want Y representable, $Y \rightarrow T$ locally closed. Now, to build Y , the claim is that it is a flattening strata for $\mathbb{P}_T^n \rightarrow T$... with respect to some coherent sheaf on $\mathbb{P}_T^n$ (which we have to build).

So, the data of $T \rightarrow \text{Gr}$ is just a vector bundle quotient

$$\pi_{T,*}F_T(d) \twoheadrightarrow V,$$

and the idea is to try to lift it to T to get a “candidate” coherent sheaf quotient $F_T \twoheadrightarrow Q$, and then Y should be the restriction of T to the part where Q is flat with correct Hilbert polynomial, i.e., the flattening stratum, which I will denote $T_{P,Q}$, which is characterized by the universal property that $T' \rightarrow T$ factors through $T_{P,Q}$ iff Q pullsback to be flat over T' with correct Hilbert polynomial.

Goal: So, we just have to build our candidate Q , and then show that $T_{P,Q}$ represents the fiber product.

3 Build Candidate Coherent Sheaf Quotient

Here's how to build Q . Call E the kernel of

$$\pi_{T,*}F_T(d) \twoheadrightarrow V,$$

giving us a short exact sequence that we can pullback to $\mathbb{P}_T^n$. This gives us

$$\begin{array}{ccccccc} 0 & \longrightarrow & \pi_T^*E & \longrightarrow & \pi_T^*\pi_{T,*}F_T(d) & \longrightarrow & \pi_T^*V \longrightarrow 0 \\ & & \searrow \alpha & & \downarrow & & \downarrow \\ & & & & F_T(d) & \longrightarrow & \text{coker}(\alpha) \longrightarrow 0 \end{array} \quad (*)$$

where $\pi_T^*\pi_{T,*}F_T(d) \rightarrow F_T(d)$ on fibers over points $t \in T$ is just the evaluate sections map. So, the morphism α is taking the “global sections” that cut out π_T^*V from $\pi_{T,*}F_T(d)$, then evaluating them. The cokernel seems like a good candidate for $Q(d)$, so define

$$Q := \text{coker}(\alpha)(-d).$$

Now, we pullback $F_T \twoheadrightarrow Q$ to the projective space over the flattening stratum $T_{P,Q}$ to get a surjection

$$F_{T_{P,Q}} \twoheadrightarrow Q_{T_{P,Q}},$$

which gives a morphism

$$T_{P,Q} \rightarrow \text{Quot}^P,$$

and we claim that this makes $T_{P,Q}$ the fiber product of T and Quot^P .

To prove the claim, we study T' points. Suppose we have

$$\begin{array}{ccc} T' & \longrightarrow & T \\ \downarrow & & \downarrow \\ \text{Quot}^P & \longrightarrow & \text{Gr} \end{array}$$

where $T' \rightarrow T$ is given by some φ , and $T' \rightarrow \text{Quot}^P$ given by a coherent sheaf quotient $F_{T'} \twoheadrightarrow Q'$ (flat over T' with correct Hilbert polynomial on fibers). The square commuting means that

$$\pi_{T',*}(F_{T'}(d) \twoheadrightarrow Q'(d)) = \pi_{T',*}F_{T'}(d) \twoheadrightarrow \varphi^*V.$$

We want to show there is a unique map to $T_{P,Q}$ commuting with the maps to T and Quot^P . By the universal property of $T_{P,Q}$, we just need to show that the constructed Q on $\mathbb{P}_T^n$ pulls back to be flat on $\mathbb{P}_{T'}^n$ with correct fibers.

This will amount to pulling back diagram $(*)$ to $P_{T'}^n$. The top row will remain exact because it consists of vector bundles, but we will also want $(\text{im}\alpha)_{T'} = \text{im}\alpha_{T'}$, which is more subtle. Study the exact sequence

$$0 \rightarrow \ker \alpha \rightarrow \pi_T^*E \rightarrow \text{im}\alpha \rightarrow 0,$$

hmmmmmm.

where I can argue that $\text{im}\alpha_{T'}$ is exactly the kernel of

$$F_T(d) \twoheadrightarrow Q'(d),$$

so the cokernel will have to just be $Q'(d)$, which is assumed to be flat over T' with the right fibers. And then, we can argue about $\text{im}\alpha_{T'}$ by

$$\mathbb{P}^n(\varphi)^*\pi_T^*V = \pi_{T'}^*\varphi^*V = \pi_{T'}^*Q'(d)$$

Question. How do I show that my candidate coherent sheaf quotient

$$F_T(d) \twoheadrightarrow Q(d)$$

will pushforward to my original vector bundle quotient

$$\pi_{T,*}F_T(d) \twoheadrightarrow V.$$

Attempts. It seems to me that the isomorphism $\pi_{T,*}F_T(d) \twoheadrightarrow V$ should be given by pushing forward the right most vertical map in my diagram, i.e., take

$$\pi_T^*V \rightarrow \text{coker}(\alpha) = Q(d),$$

then push it forward

$$V \cong \pi_{T,*}\pi_T^*V \rightarrow \pi_{T,*}Q(d),$$

where this isomorphism is given by the projection formula and the fact that $\pi_{T,*}\mathcal{O}_{\mathbb{P}_T^n} \cong \mathcal{O}_T$.

So, I have a natural map... I just need to see that it is an isomorphism.

Fact 1. The morphism $\pi_T^*V \rightarrow Q(d)$ is surjective, since in the square

$$\begin{array}{ccc} \pi_T^*\pi_{T,*}F_T(d) & \longrightarrow & \pi_T^*V \\ \downarrow & & \downarrow \\ F_T(d) & \longrightarrow & Q(d) \end{array}$$

the left vertical map and the bottom horizontal map are surjective (the vertical one by global generation and regularity of the fibers of $F_T(d)$ over T , and the horizontal one is projection to a cokernel). So, if I could pin down the kernel J of

$$\pi_T^*V \rightarrow Q(d),$$

I could have success by showing

- $\pi_{T,*}J = 0$
- $R^1\pi_{T,*}J = 0$

which I could maybe accomplish by showing

$$H^i(\mathbb{P}_{k(t)}^n, J|_{\mathbb{P}_{k(t)}^n}) = 0$$

for $i = 0, 1$. But, to do this I need a better hold on J ... I would like to understand the fibers of J by restricting the surjection $\pi_T^*V \rightarrow Q(d)$ to fibers, but $Q(d)$ need not be flat (so I need not get the exactness I want, since the inclusion of a fiber is very much not flat in general).